

SOC-5811 Week 9: Potential outcomes

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REVIEW

Research Question: What is the difference in grandchild wealth by grandparent race, controlling for grandparent wealth?

- ▶ Y = grandchild wealth (continuous)
- ▶ X = grandparent wealth (continuous)
- ▶ Z = grandparent race (categorical: Black, White)

A: $y_i = \beta_0 + \boxed{\beta_1}(X)$

B: $y_i = \beta_0 + \boxed{\beta_1}(X) + \beta_2(Z)$

C: $y_i = \beta_0 + \beta_1(X) + \boxed{\beta_2}(Z)$

D: $y_i = \beta_0 + \beta_1(X) + \beta_2(Z) + \boxed{\beta_3}(X \times Z)$

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Research Question: What is the difference in the average association between grandparent wealth and grandchild wealth by race?

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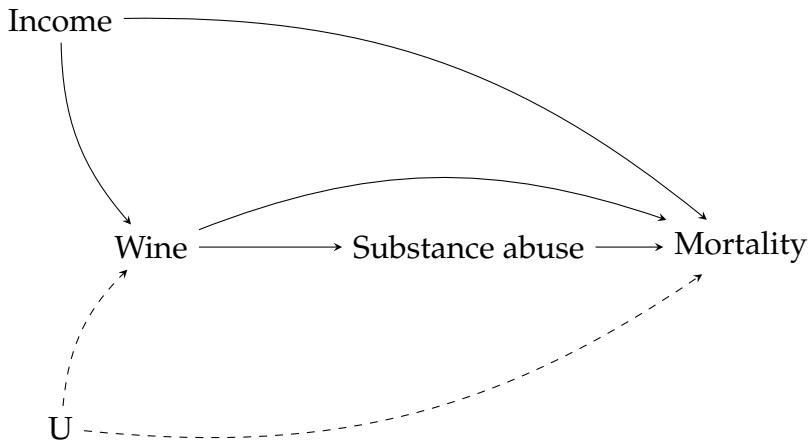
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Key steps for building a quantitative research question:

- ▶ **(1)** Draw a conceptual model to guide your research question.



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REVIEW DAGs

Key steps for building a quantitative research question:

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- ▶ **(2)** Read the literature. What do we know?
- ▶ **(3)** Refine conceptual model into a DAG (this may include drawing the “causal field”).
- ▶ **(4)** Think systematically and explicitly about the potential effect of bias.



DAGs

Let's workshop a DAG on the board for someone's research project.



CAUSAL INFERENCE

- ▶ Causal inference involves counterfactuals.



CAUSAL INFERENCE

- ▶ Causal inference involves counterfactuals.
- ▶ What is a *counterfactual*?



POTENTIAL OUTCOMES

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POTENTIAL OUTCOMES

We observe some value of y for every unit i :

$$y_i$$



POTENTIAL OUTCOMES

“The fundamental problem of causal inference”

$$\delta_i = y_i^1 - y_i^0$$



POTENTIAL OUTCOMES

“The fundamental problem of causal inference”

$$\delta_i = y_i^1 - y_i^0$$

- ▶ Experiments do not overcome this problem!
- ▶ The true experiment would be the same individual at the same time.



POTENTIAL OUTCOMES

Group	Y1	Y0
Treatment ($X=1$)	Observable	Counterfactual
Control ($X=0$)	Counterfactual	Observable

POTENTIAL OUTCOMES

Group	Y1	Y0
Treatment (X=1)	$y_i^1 \mid X = 1$	$y_i^0 \mid X = 1$
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POTENTIAL OUTCOMES

Group	Y1	Y0
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- ▶ Counterfactuals exist in theory but are **unobservable**.
- ▶ **Causal inference is a prediction problem!**

AVERAGE TREATMENT EFFECTS

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- ▶ We can't predict individual treatment effects, but we can predict the **average treatment effect (ATE)**.
- ▶ The regression model is one way of generating the **predictions** needed to estimate the ATE.

CAUSAL INFERENCE: PREDICTION PROBLEM

$$\text{prestige}_i = \beta_1 \text{college}_i + \beta_2 \text{age}_i + \beta_3 \text{race}_i + \beta_4 \text{sex}_i + \beta_5 \text{sibs}_i + \beta_6 \text{fpres}_i$$

```
##
## Call:
## lm(formula = prestige10 ~ college + age + as.factor(race) + sex +
##     sibs + papres10, data = gss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -40.745  -8.049  -0.452   7.906  37.196
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   35.48531    1.37482   25.811 < 2e-16 ***
## college       13.49147    0.46057   29.293 < 2e-16 ***
## age           0.08419    0.01315    6.400 1.8e-10 ***
## as.factor(race)2 -1.71212    0.79528   -2.153 0.03141 *
## as.factor(race)3 -0.27320    0.79941   -0.342 0.73257
## sex          -0.68775    0.44159   -1.557 0.11948
## sibs         -0.26907    0.08859   -3.037 0.00241 **
## papres10       0.06071    0.01760    3.449 0.00057 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.82 on 2917 degrees of freedom
## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2765
## F-statistic: 160.6 on 7 and 2917 DF, p-value: < 2.2e-16
```


CAUSAL INFERENCE: PREDICTION PROBLEM

##	college	prestg10	y0	y1
##	<num>	<num>	<num>	<num>
## 1:	0	38	40.33558	53.82706
## 2:	1	57	43.68202	57.17349
## 3:	1	40	35.71267	49.20415
## 4:	0	60	37.58241	51.07388
## 5:	1	60	40.80190	54.29338
## 6:	0	16	37.78251	51.27398

CAUSAL INFERENCE: PREDICTION PROBLEM

```
mean(gss$y1)
```

```
## [1] 54.17606
```

```
mean(gss$y0)
```

```
## [1] 40.68459
```

CAUSAL INFERENCE: PREDICTION PROBLEM

```
mean(gss$y1) - mean(gss$y0)
```

```
## [1] 13.49147
```

CAUSAL INFERENCE: PREDICTION PROBLEM

```
mean(gss$y1) - mean(gss$y0)
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```
## [1] 13.49147
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```
summary(mod)
```

```
##
## Call:
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- ▶ We can't predict individual treatment effects, but we can predict the **average treatment effect (ATE)**. The regression model is one way of generating the **predictions** needed to estimate the ATE.
- ▶ But when can these **predictions** be interpreted as **potential outcomes**?

AVERAGE TREATMENT EFFECTS

Independence assumption

$$(Y^1, Y^0, \delta) \perp\!\!\!\perp X$$

- ▶ Also referred to as exchangeability, no confounding, etc.

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When will all three of these be identical?

REVIEW

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- ▶ What are potential outcomes?
- ▶ What is the fundamental problem of causal inference?
- ▶ What is the independence or exchangeability assumption?



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- ▶ Panel data involves repeated observations for the same unit. How is this possible?
- ▶ Same units observed at different **times**.



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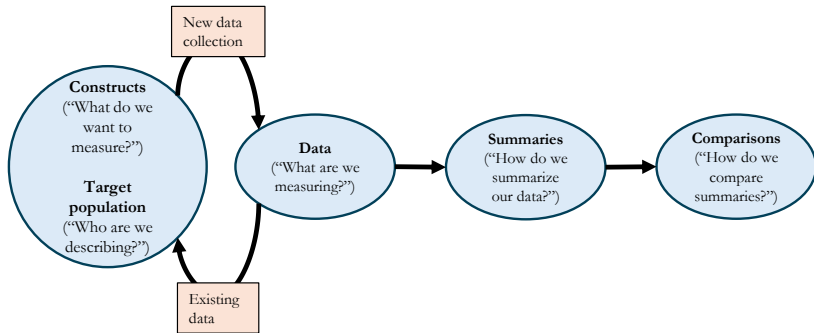
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- Interpreting model parameters as causal effects: DAGs, assumptions, etc.

GENERALIZATION: SUMMARY

What are we generalizing?

1. Measurement.
2. Generalizing from sample to population.
3. Comparison (e.g., causal inference).
4. Forecasting.





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- ▶ **Statistical inference:** We observe data and we try to learn something from it, which involved uncertainty. Based on our sample, how sure are we that the population parameter is different from 0?
- ▶ **Sampling distribution** = the set of possible datasets that could have been observed if the data collection process had been re-done, along with the probabilities of these values.



INFERENCE

- ▶ **Standard error** = a measure of the spread of sample means around the population mean; the standard deviation of the sampling distribution.

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- ▶ **p-value** = the probability that the null model would produce a parameter estimate as extreme as your observed estimate.
- ▶ **Statistical significance** = conventionally defined as a p-value less than 0.05, relative to some null hypothesis or prespecified value that would indicate no effect present.



INFERENCE

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4. STATISTICAL INFERENCE

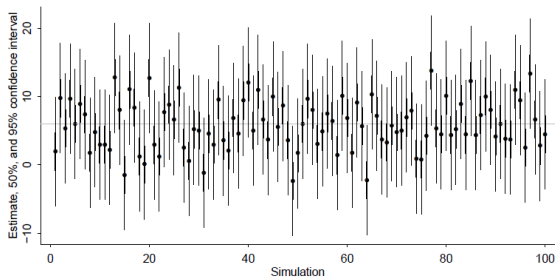


Figure 4.2. Simulation of coverage of confidence intervals: the horizontal line shows the true parameter value, and dots and vertical lines show estimates and confidence intervals obtained from 100 random simulations from the sampling distribution. If the model is correct, 50% of the 50% intervals and 95% of the 95% intervals should contain the true parameter value, in the long run.

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INFERENCE

- ▶ How do we estimate standard errors?
 - ▶ Analytic closed-form solution.



- ▶ How do we estimate standard errors?
 - ▶ Analytic closed-form solution.
 - ▶ Simulate the sampling distribution using the sample data (“bootstrapping”).

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- ▶ H_1 : There is some relationship between Income and Education.

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$$\text{Income}_i = \beta_0 + \boxed{\beta_1} \text{Edu} + \beta_2 \text{Age} + \epsilon_i$$

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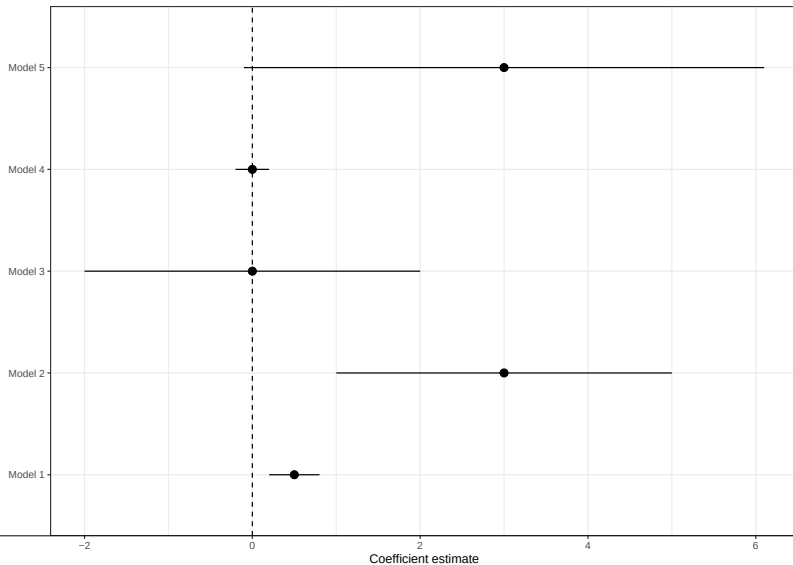
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- ▶ H_0 : There is no relationship between Income and Edu, controlling for Age.
- ▶ H_1 : There is some relationship between Income and Edu, controlling for Age.

- ▶ What does a small p-value mean? Does a small p-value mean that the observed association between X and Y is *important*?

INFERENCE



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MODELS

- ▶ What is a model? A combination of *deterministic* and *stochastic* processes.



MODELS

- ▶ What is a model? A combination of *deterministic* and *stochastic* processes.
- ▶ What does it mean to “fit” a regression model? In other words, once we have written down a model, how do we estimate it?



MODELS

1. **Estimand:** What is my goal? What is my target comparison I want to infer something about? Example: I want to know if income varies by education, conditional on age.
2. **Estimator:** How do I relate my population estimand to real sample data in a way that let's me identify its *most likely* value given that sample data?
3. **Estimation:** How do I optimize my estimator (i.e., calculate the most likely value of my estimand)?



MODELS

1. **Estimand:** Choose a functional form for the model and highlight the target parameter (“estimand”).

$$y_i = \beta_0 + \boxed{\beta_1} x_i + \epsilon_i$$

2. **Estimator:** Choose an estimator that relates the model to real observed data (e.g., OLS) and is unbiased, consistent, and efficient.

$$\text{Sum of squared residuals} = \sum_{i=1}^n (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$$

3. **Estimation:** Estimate parameters by optimizing the estimator; sometimes this can be done analytically, sometimes it requires brute computational force.

COEFFICIENTS

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- ▶ How do we interpret β_1 if X is continuous (e.g., age)?
- ▶ How do we interpret β_1 if X is categorical (e.g., level of education)?
- ▶ When might we want to *rescale* or *shift* X? How does that change our interpretation of β_1 ?



COEFFICIENTS

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- ▶ **Adjusted comparisons:** Sometimes comparisons at a high level can be misleading about comparisons at a lower level (Simpson's Paradox). Example: USA and Mexico mortality rates.
 - ▶ Am I doing causal inference? No, I'm just comparing rates conditional on age.
- ▶ **Causal inference:** Confounding.

COEFFICIENTS

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- How do I interpret β_1 ?

COEFFICIENTS

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- ▶ How do I interpret β_1 ?
- ▶ Does the estimate of β_1 change if I rescale or shift age?

Categorical $\times$ Continuous

$$\text{Income}_i = \beta_0 + \beta_1 \text{College}_i + \beta_2 \text{Age}_i + \beta_3 (\text{College}_i \times \text{Age})_i$$

- How do I interpret β_1 ?

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- How do I interpret β_2 ?

Categorical $\times$ Continuous

$$\text{Income}_i = \beta_0 + \beta_1 \text{College}_i + \beta_2 \text{Age}_i + \beta_3 (\text{College}_i \times \text{Age})_i$$

- ▶ How do I interpret β_2 ?
- ▶ How do I interpret β_3 ?

Categorical $\times$ Continuous

$$\text{Income}_i = \beta_0 + \beta_1 \text{College}_i + \beta_2 \text{Age}_i + \beta_3 (\text{College}_i \times \text{Age})_i$$

- How many possible predictions can we make from this model?

COEFFICIENTS

Continuous $\times$ Continuous

$$\text{Income}_i = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Income}_i^{\text{parent}} + \beta_3 (\text{Age} \times \text{Income}_i^{\text{parent}})$$

- How many possible predictions can we make from this model?

COEFFICIENTS

Categorical $\times$ Categorical

$$\text{Income}_i = \beta_0 + \beta_1 \text{College}_i + \beta_2 \text{Immigrant}_i + \beta_3 (\text{College}_i \times \text{Immigrant}_i)$$

- How many possible predictions can we make from this model?

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$$\text{Income}_i = \beta_0 + \beta_1 \text{College}_i + \beta_2 \text{Immigrant}_i + \beta_3 (\text{College}_i \times \text{Immigrant})_i$$

- ▶ Non-college, non-immigrant = β_0
- ▶ College, non-immigrant = β_1
- ▶ Non-college, immigrant = β_2
- ▶ College, immigrant = $\beta_1 + \beta_2 + \beta_3$

CAUSAL INFERENCE

From *Regression and Other Stories* (6.3): “The safest interpretation of a regression is as a comparison ... Under some conditions, the between-person inferences from a regression analysis can be interpreted as causal effects, but as a starting point we recommend describing regression coefficients in predictive or descriptive, rather than causal, terms.

To summarize: regression is a mathematical tool for making predictions. Regression coefficients can **sometimes** be interpreted as effects, but they can **always** be interpreted as average comparisons.”



CAUSAL INFERENCE

- ▶ What is the key assumption to interpret average comparisons as causal effects?



CAUSAL INFERENCE

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- ▶ Exchangeability (or the independence assumption).



CAUSAL INFERENCE

When can we make strong claims about this assumption? What are our strategies?

- ▶ Randomization of the exposure.



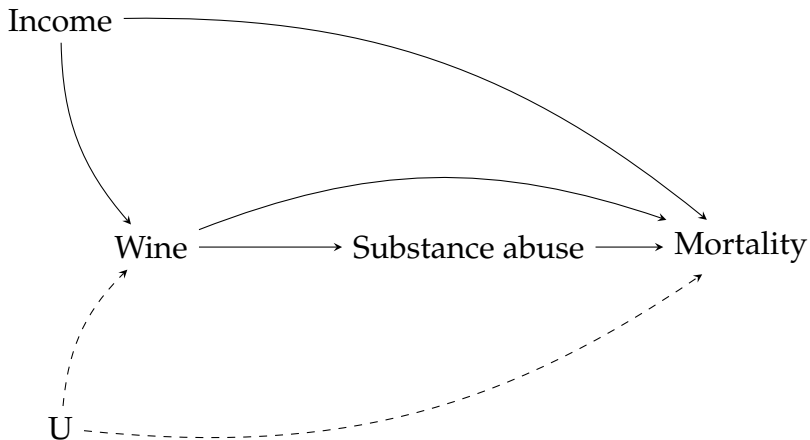
CAUSAL INFERENCE

When can we make strong claims about this assumption? What are our strategies?

- ▶ Randomization of the exposure.
- ▶ Controlling for confounders.



CAUSAL INFERENCE



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- ▶ We can't predict individual treatment effects, but we can predict the **average treatment effect (ATE)**.
- ▶ Causal inference is a prediction problem: we must predict potential outcomes using average observed outcomes from exchangeable units.
- ▶ The regression model is one way of generating the **predictions** needed to estimate the ATE.

CAUSAL INFERENCE

Group	Y1	Y0
Treatment ($X=1$)	Observable	Counterfactual
Control ($X=0$)	Counterfactual	Observable

LOOKING AHEAD

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- ▶ For the rest of the semester, we will introduce a few other advanced uses of the basic linear regression model.
- ▶ In SOC 8811, you will learn about more complex regression models (e.g., logistic regression).

